

Theory of Automata

Assignment 3

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Q1:

(a)

$$S \rightarrow aSbb \mid aSbbb \mid abb \mid abbb$$

(b)

$$S \rightarrow abSaX \mid \lambda$$

$$X \rightarrow bX \mid b$$

(c)

$$S \rightarrow aA \mid bS \mid \lambda$$

$$A \rightarrow aA \mid bB \mid \lambda$$

$$B \rightarrow aA \mid bC \mid \lambda$$

$$C \rightarrow bS \mid \lambda$$

(d)

$$S \rightarrow XXX$$

$$X \rightarrow a \mid b$$

or

$$S \rightarrow aX_1 \mid bX_1$$

$$X_1 \rightarrow aX_2 \mid bX_2$$

$$X_2 \rightarrow a \mid b$$

Q2:

(a)

$$\text{Rem } C \rightarrow \lambda:$$

$$S \rightarrow aC \mid bC \mid a \mid b$$

$$C \rightarrow a \mid b$$

(b)

$$\text{Rem } C \rightarrow \lambda:$$

$$S \rightarrow aC \mid bS \mid a \mid b$$

$$C \rightarrow aC \mid a$$

(c)

$$\text{Rem } Y \rightarrow \lambda:$$

$$S \rightarrow YaY \mid bY \mid a \mid b \mid Yc \mid aY$$

$$Y \rightarrow YaY \mid YbY \mid a \mid b \mid Ya \mid aY \mid Yb \mid bY$$

Q3:

(a)

Rem $C \rightarrow D$!
 $S \rightarrow CD$
 $C \rightarrow aD / Db / \lambda$
 $D \rightarrow aD / Db / \lambda$

(b)

Rem $A \rightarrow B$!
 $S \rightarrow AA$
 $A \rightarrow aB / Bb / \lambda / BB$
 $B \rightarrow aB / Bb / \lambda$

Q4:

(a)

$S \rightarrow AX / SX / a$
 $X \rightarrow SA$
 $A \rightarrow a$

(b)

$S \rightarrow XB / c$
 $B \rightarrow b$
 $A \rightarrow a$
 $C \rightarrow AC$ Remove
 $X \rightarrow AA$

$S \rightarrow XB / c$

$B \rightarrow b$

$A \rightarrow a$

$X \rightarrow AA$

(c)

$S \rightarrow aS / Z$

$Z \rightarrow a / b$

$C \rightarrow b$ Remove

$S \rightarrow aS / a / b$

$Z \rightarrow a / b$ Rem

$S \rightarrow aS / a / b$

(d)

$S \rightarrow SX_1 / SX_3 / SX_4 / \lambda$

$X_1 \rightarrow AS$

$X_2 \rightarrow BS$

$X_3 \rightarrow X_1 X_2$

$X_4 \rightarrow X_2 X_1$

$A \rightarrow a$

$B \rightarrow b$

Q5:

(a)

$$\begin{array}{lcl}
 S \rightarrow AX \mid SX \mid a & \xrightarrow{\text{Rem } S \rightarrow SX} & S \rightarrow AXS' \mid aS' \mid AX \mid a \\
 X \rightarrow SA & & S' \rightarrow XS' \mid X \\
 A \rightarrow a & & X \rightarrow SA \\
 & & A \rightarrow a
 \end{array}$$

Put a in A, viceversa

$$\begin{array}{lcl}
 S \rightarrow aXS' \mid aS' \mid aX \mid a & & S \rightarrow aXS' \mid aS' \mid aX \mid a \\
 S' \rightarrow XS' \mid X & & S' \rightarrow XS' \mid X \\
 X \rightarrow SA & & X \rightarrow aXS'A \mid aS'A \mid aXA \mid aA \\
 A \rightarrow a & & A \rightarrow a
 \end{array}$$

$$\begin{array}{l}
 S \rightarrow aXS' \mid aS' \mid aX \mid a \\
 S' \rightarrow aXS'A S' \mid aS'AS' \mid aXAS' \mid aAS' \mid aXS'A \mid aS'A \mid aXA \mid aA \\
 X \rightarrow aXS'A \mid aS'A \mid aXA \mid aA \\
 A \rightarrow a
 \end{array}$$

(b)

$$\begin{array}{lcl}
 S \rightarrow XB \mid c & S \rightarrow XB \mid c & S \rightarrow aAB \mid c \\
 B \rightarrow b & B \rightarrow b & B \rightarrow b \\
 A \rightarrow a & A \rightarrow a & A \rightarrow a \\
 X \rightarrow AA & X \rightarrow aA & X \rightarrow aA \text{] Remove}
 \end{array}$$

$$\begin{array}{l}
 S \rightarrow aAB \mid c \\
 B \rightarrow b \\
 A \rightarrow a
 \end{array}$$

(c)

$$\begin{array}{lcl}
 S \rightarrow AS \mid a \mid b & \longrightarrow & S \rightarrow aS \mid a \mid b \\
 A \rightarrow a & & \text{A} \rightarrow a \text{ Remove}
 \end{array}$$

(d)

$$S \rightarrow SX_1 \mid SX_2 \mid SX_4 \mid \lambda$$

$$X_1 \rightarrow AS$$

$$X_2 \rightarrow BS$$

$$X_3 \rightarrow X_1X_2$$

$$X_4 \rightarrow X_2X_1$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$S \rightarrow SX_1 \mid SX_3 \mid SX_4 \mid \lambda$$

$$X_1 \rightarrow aS$$

$$X_2 \rightarrow bS$$

$$X_3 \rightarrow aSX_2$$

$$X_4 \rightarrow bSX_1$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$S \rightarrow S' \mid \lambda$$

$$S' \rightarrow X_1S' \mid X_3S' \mid X_4S' \mid X_1 \mid X_3 \mid X_4$$

$$X_1 \rightarrow aS$$

$$X_2 \rightarrow bS$$

$$X_3 \rightarrow aSX_2$$

$$X_4 \rightarrow bSX_1$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$S \rightarrow aSS' \mid aSX_2S' \mid bSX_1S' \mid aS \mid aSX_2 \mid bSX_1 \mid \lambda$$

$$S' \rightarrow aSS' \mid aSX_2S' \mid bSX_1S' \mid aS \mid aSX_2 \mid bSX_1$$

$$X_1 \rightarrow aS$$

$$X_2 \rightarrow bS$$

$$X_3 \rightarrow aSX_2$$

$$X_4 \rightarrow bSX_1$$

$$A \rightarrow a$$

$$B \rightarrow b$$